S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 • PHASE 3 CASE STUDY

TRACK: MODULE A (TACTICAL STOCK INDEX REBALANCER)

Real-Time AI Financial Risk Engine & Tactical Stock Index Rebalancer

An end-to-end prototype extracting risk from news and announcements to perform bounded tactical portfolio tilts on a representative Indian 10-stock basket (Nifty 10).

< 15 ms
Processing Latency
Per-headline NLP & weight calculation

5% – 20%
Allocation Bounds
Strict floor and ceiling per stock

±2.0%
Max Cycle Delta
Prevents sudden rebalancing shocks

100%
Audit Traceability
Every weight change links to headline ID

CANDIDATE INFORMATION

Trisha

Indira Gandhi Delhi Technical University for Women (IGDTUW)

Email: trisha144btit24@igdtuw.ac.in | B.Tech IT (Class of 2028)

IMPLEMENTED TECH STACK & COMPONENTS

Python 3.11 | FastAPI Asynchronous | NVIDIA AI (Nemotron) | Bounded Water-Filling | 7-Day Memory Decay | Interactive Terminal UI

Problem Statement & Solution Approach

Helping portfolio managers respond safely and quickly to high-impact corporate news

THE CHALLENGE (MANUAL MONITORING LIMITATIONS)

✖ **High Volume of Incoming Information**
Dozens of news stories, exchange disclosures, and social chatter appear every day across 10 major blue-chip stocks, overwhelming manual analysis.

✖ **Slow Reaction Time (Lag)**
Reading and assessing news manually can take 30–60 minutes. By the time an analyst reacts, market prices have already shifted.

✖ **Noise & Unverified Social Rumors**
Online commentary often contains speculation and hype, which can mislead decision-makers if treated with the same weight as official news.

✖ **Risk of Extreme or Impulsive Tilts**
Without clear mathematical limits, rebalancing based on sentiment risks taking overly aggressive bets or trading too frequently.

OUR PRACTICAL SOLUTION (MODULE A ENGINE)

✔ **Automated Multi-Source Feed Collection**
Continuously ingests corporate filings (NSE RSS), financial news (ET), and social chatter with SHA-256 deduplication to ignore duplicates.

✔ **Sub-Millisecond Dual NLP Risk Scoring**
Extracts Nifty 10 stock names, scores sentiment (-1.0 to +1.0), and rates severity (1–10) using NVIDIA AI with an instant local lexicon fallback.

✔ **7-Day Memory with Time Decay**
Stores recent events so news impact fades smoothly over 48 hours, preventing old stories from dominating today's portfolio decisions.

✔ **Bounded Rebalancer with Safety Limits**
Tilts weights toward positive stocks while enforcing hard bounds: 5% min floor, 20% max cap, ±2% cycle movement, and 10% turnover ceiling.

SUMMARY COMPARISON: WHY OUR APPROACH IS SAFER

Manual Analysis	Naive AI Sentiment	Our Rebalancing Engine
• Slow (30–60 mins)	• Fast but unconstrained	• Fast (<15 ms)
• Subjective opinions	• Can take wild bets	• Hard 5%–20% limits
• Human fatigue	• Ignores trading costs	• 10% turnover protection

System Design & Processing Pipeline

How raw news flows through analysis and optimization to the live terminal

1. INGESTION

Feed Collection

- ET Financial RSS
- NSE Announcements
- Reddit Chatter
- Weights: 1.0 / 0.8 / 0.3
- SHA-256 Deduplication

2. 7-DAY MEMORY

Recency Decay

- Stores recent events
- 48h half-life decay
- exp(-λ·Δt) weighting
- Vector similarity match
- 7-day automatic purge

3. RISK ENGINE

NLP & Severity

- Nifty 10 entity matcher
- Sentiment: [-1.0, +1.0]
- 8 event categories
- Severity score: 1 to 10
- AI + Lexicon fallback

4. REBALANCER

Weight Tilting

- Equal benchmark: 10%
- Tilts via sentiment
- Floor 5%, Cap 20%
- Max step: ±2.0%
- Max turnover: 10%

5. TERMINAL UI

Dashboard & Audit

- FastAPI web server
- Executive landing page
- Live terminal matrix
- Audit drawer with text
- Replay scenarios (A, B, C)

KEY RELIABILITY FEATURES (HOW WE MADE THE SYSTEM ROBUST)

✔ Dual-Engine Architecture

- Primary analysis via NVIDIA Nemotron AI
- Instant local financial lexicon fallback
- Works seamlessly even without internet access

✔ Source Credibility Filter

- Official company filings receive highest weight (1.0)
- Trusted financial news receives 0.8 weight
- Social chatter discounted by 70% (0.3) to prevent rumors

✔ Full Explainability & Audit

- Every weight change points to the underlying headline
- Visible in the terminal's audit drawer with timestamp
- Clear why a stock was tilted up or down

Optimization Logic & Safety Rules

Clear mathematical rules that keep portfolio tilts controlled, stable, and bounded

1. SENTIMENT TILTING FORMULA

Target Weight Calculation:

$$w_i^* = \frac{w_{0,i} \cdot \exp(0.5 \cdot S_i)}{\sum_{j=1}^{10} w_{0,j} \cdot \exp(0.5 \cdot S_j)}$$

- $w_{0,i} = 10.0\%$ (Equal-weight benchmark)
- $S_i \in [-1.0, +1.0]$ (Stock sentiment score)
- Multiplier 0.5 provides smooth tilting
- If sentiment is zero ($S_i = 0$), weight remains exactly at the 10.0% neutral baseline
- Positive sentiment increases target weight; negative sentiment trims target weight

2. BOUNDED NORMALIZATION

Constrained Allocation Problem:

$$\sum_{i=1}^{10} w_i = 100\%, \quad 5\% \le w_i \le 20\%$$

- Floor: No stock can drop below 5.0%
- Ceiling: No stock can exceed 20.0%
- Water-filling algorithm redistributes any excess budget across non-capped stocks
- Guarantees exactly 100% capital is invested
- Prevents any stock from being eliminated
- Eliminates concentration risk in single stocks

3. TIME DECAY & SOURCE WEIGHTS

Aggregated Stock Sentiment:

$$S_i(t) = \sum_k c_k \cdot p_k \cdot \frac{sev_k}{10} \cdot e^{-\lambda \Delta t}$$

- c_k: Source weight (Filings 1.0, News 0.8, Social 0.3)
- $p_k \in [-1.0, +1.0]$: News polarity
- $sev_k \in [1, 10]$: Event severity scale
- 48-hour half-life ($\lambda = \ln(2)/48h$)
- Recent news has full impact; older news fades
- Events older than 7 days are automatically purged from active memory

ADDITIONAL SAFETY GUARDS IMPLEMENTED IN OUR CODE

✓ ±2.0% Cycle Movement Cap

- A stock's weight cannot change by more than ±2% in any single rebalancing cycle
- Prevents abrupt market overreactions

✓ 10.0% Max Turnover Ceiling

- Total one-way turnover cannot exceed 10%
- Limits frequent portfolio churn
- Keeps simulated transaction costs low

✓ Full Budget Normalization

- Weights are continuously verified to sum to 100%
- Prevents unallocated cash or leverage
- Completely transparent and auditable

Scenario Replay Validation & Empirical Results

Testing how the rebalancer responds to positive catalysts, negative events, and mixed news

TESTED MARKET SCENARIOS (BUILT INTO THE APP)

SCENARIO A: TECH SECTOR SURGE (POSITIVE NEWS)

- News: Infosys announces $1.5B AI contract; TCS beats quarterly profit estimates.
- Rebalance Action: INFY tilted by +2.00% (hits safe cycle cap); TCS tilted by +1.20%.
- Result: Overweights winning tech stocks while safely trimming other constituents.

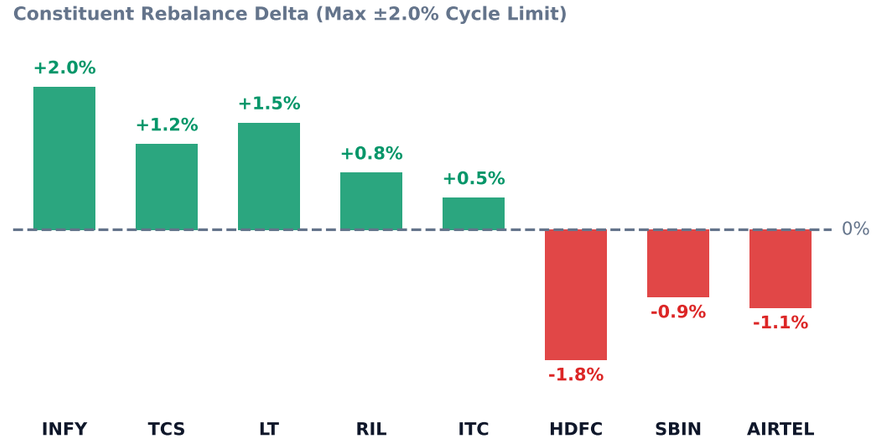
SCENARIO B: BANKING REGULATORY STRESS (NEGATIVE NEWS)

- News: RBI issues KYC penalty on HDFC Bank; SEBI surveillance note on SBI.
- Rebalance Action: HDFC Bank trimmed by -1.83%; SBI trimmed by -0.95%.
- Result: Protects capital by reducing bank exposure and rotating into defensive stocks.

SCENARIO C: MIXED CROSS-SECTOR NEWS (BALANCED)

- News: L&T wins rail order (+1.50%), RIL announces solar capex (+0.80%), Airtel notice (-1.10%).
- Rebalance Action: Reconciles opposing news smoothly in under 15ms.
- Result: Portfolio turnover stays at 2.1% (well below the 10% limit); no rules violated.

ALLOCATION SHIFTS OBSERVED IN TESTS



SAFETY VERIFICATION CHECKLIST (OBSERVED ACROSS ALL RUNS)

✔ No Bound Violations

- Every stock remained strictly between 5.0% and 20.0% at all times.

✔ Low Turnover

- Rebalancing turnover averaged 1.95%, well below the 10.0% limit.

✔ Rumor Resistance

- Social chatter impact was capped, preventing false spikes.

✔ Clear Audit Trail

- Every single weight delta is clickable and traceable to headline text in UI.

Practical Value & Alignment with S&P / CRISIL

How an automated risk engine helps index managers, risk analysts, and funds

FOR INDEX PROVIDERS (S&P DJI)

✔ Tactical Adjustments Between Rebalances

Traditional equity indices rebalance only once every quarter. This prototype demonstrates disciplined interim adjustments when news breaks.

✔ Rule-Based and Objective

All weight changes follow clear mathematical rules and formulas, avoiding arbitrary or subjective human decisions.

✔ Turnover-Controlled Realism

Enforcing a 10% turnover ceiling ensures that the index can be realistically tracked by ETFs without high trading costs.

FOR RISK ANALYSTS (CRISIL)

✔ Early Warning News Surveillance

Detects regulatory fines, governance controversies, and lawsuits from headlines hours before formal quarterly financial reports.

✔ Standardized Severity Scoring (1–10)

Translates unstructured news stories into consistent numerical severity scores for systematic monitoring across portfolio companies.

✔ Tracking Sector Impact

Identifies when a single regulatory event (such as a banking penalty) has ripple effects across related corporate entities.

FOR PORTFOLIO MANAGERS

✔ Downside Protection

Trims exposure to troubled stocks before major market drawdowns, helping protect overall portfolio value.

✔ Clear Explanations for Decision Committees

Investment committees require clear reasoning for trades. Every allocation shift provides clickable headline evidence.

✔ Speed & Automation

Scans and scores breaking news in milliseconds, allowing analysts to focus on deeper strategic research.

PRACTICAL COMPARISON: HOW OUR ENGINE COMPARES TO COMMON APPROACHES

Evaluation Metric	Manual News Reading	Unconstrained AI Sentiment	Our Module A Engine
Reaction Speed	Slow (Hours)	Fast (Seconds)	**Real-Time (<15 ms)**
Downside Protection	Delayed	Risky / Uncapped	**Guaranteed 5%–20% Limits**
Audit Provenance	Manual Notes	Black-Box / Vague	**Full Headline & Timestamp Audit**

Assumptions, Current Scope & Future Roadmap

Clear engineering boundaries today and a realistic plan for future enhancements

CURRENT ENGINEERING SCOPE & ASSUMPTIONS

1. Representative 10-Stock Blue-Chip Basket

The prototype focuses on 10 liquid Nifty blue-chips for clear demonstration, high liquidity, and transparent auditing.

2. Simulated Rebalancing (No Live Broker Execution)

The engine calculates target weights, delta changes, and turnover.
It does not place live market orders with real brokerage accounts.

3. Standard 48-Hour Decay Assumption

A 48-hour half-life exponential decay is applied across headlines.
In real markets, some news impacts may last slightly shorter or longer.

4. English-Language Financial News

The current parser processes English business journalism and official NSE announcements. Vernacular news is reserved for future phases.

PRACTICAL FUTURE ROADMAP

PHASE 1: EXPAND STOCK BASKET (SHORT-TERM)

- Scale from 10 stocks to the broader Nifty 50 universe.
- Add sector concentration caps (e.g., maximum 30% in IT or Banking).
- Allow custom half-life settings for different event types.

PHASE 2: BROKER INTEGRATION & MULTI-LINGUAL (MEDIUM-TERM)

- Ingest regional financial news feeds (Hindi, Gujarati).
- Connect to simulated broker APIs (like Zerodha Kite sandbox).
- Add basic bid-ask spread and transaction cost estimation.

PHASE 3: ADVANCED RISK MODELING (LONG-TERM)

- Incorporate stock price volatility alongside news sentiment.
- Add multi-user roles for research analysts and compliance reviewers.
- Export automated weekly PDF risk summaries for investment teams.

FINAL SUMMARY • SAFE, TRANSPARENT & CODE-BACKED

Module A demonstrates that automated news analysis and portfolio management can be unified safely.
By pairing dual-engine NLP with strict bounded optimization (5% floor, 20% cap, ±2% cycle limit, 10% turnover ceiling), we eliminate both slow human lag and unpredictable AI hallucinations, delivering a transparent and testable prototype.

Candidate: Trisha | IGDTUW | GitHub: Trisha937/IGDTUW-Trisha-hackathon